

2.2.5 Burning / Recording

Burning refers to the process of recording data on to Recordable media. It is called burning since the heat of the laser beam causes the dye to change from transparent to opaque. The opaque areas do not reflect light as well as the transparent area, thus simulating the behaviour of actual pits and lands.

2.2.6 CAV

Constant Angular Velocity refers to the rotational characteristic of the drive. In CAV mode, the rpm remains the same, so data closer to the outer margins of the CD are read or written faster than when present on the inner side. Most drives use this mode of operation when dealing with data nearer to the inner margin, and switch to CLV when closer to the outer margin (see next).

2.2.7 CLV

Constant Linear Velocity drives do not have a fixed rpm, but change this to allow the same length of track to be covered. This is made possible by using a higher rpm when reading areas close to the inner side, and using a lower rpm when reading data off the outer side.

2.2.8 CD

The Compact Disc was the first popular optical storage media. The result of a collaboration between Philips and Sony, its original use was for playback of audio recordings. CDs are available in the following types based on the content they carry: CD A (Audio CD), Video CD (for video), Mixed Mode CD (Containing audio and other data types), Data CD (containing all forms of files), CD-I (CDs containing interactive content), and more. Based on their recording capability, we have CD-ROM (Read-only; all pre-manufactured CDs are of this type); CD-R (recordable once); and CD-RW (read and write many times). The original CD drive could read data at 150 Kilobytes per second, and every subsequent CD drive speed has been rated as a multiple of this speed. The laser used in a CD drive has a wavelength of 780 nanometres. The rated storage capacity is 700 MB.